Brainstorming & Idea Prioritization Template

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	3 Marks

Project Title

Personalized Travel Itinerary Planner

Brainstorming Session: Feature Ideas

Core Features (Must-Have)

1. **AI-Powered Itinerary Generation** — Use Gemini / Groq LLMs to generate complete, multi-day itineraries based on user inputs
2. **User Authentication** — Secure signup/login via Supabase Auth with JWT-based session management
3. **Trip CRUD Operations** — Create, read, update, and delete trips with full metadata
4. **Day-by-Day View** — Display itinerary organized by days with activities, times, locations, and costs
5. **Budget Tracking** — Per-activity cost estimation with day totals and overall trip total
6. **Activity Management** — Edit, delete, and reorder individual activities within a day

Enhanced Features (Should-Have)

7. **Day Regeneration** — Regenerate a single day's activities using AI without affecting other days
8. **Drag-and-Drop Reordering** — Reorder activities within a day using drag handles
9. **Travel Style Support** — Three modes: relaxed (3 activities/day), balanced (4-5), packed (6-7)
10. **Interest-Based Personalization** — Allow users to specify interests (history, food, adventure, etc.)
11. **Multi-Currency Support** — Accept any 3-letter currency code for budget planning
12. **Responsive UI** — Mobile-friendly design using TailwindCSS and shadcn/ui

Future Features (Nice-to-Have)

13. **Collaborative Trip Planning** — Invite trip members with viewer/editor roles
14. **Map Integration** — Display activities on an interactive map
15. **Export to Calendar** — Export itinerary to Google Calendar / iCal
16. **Booking Links** — Direct links to book hotels, restaurants, attractions
17. **Weather Integration** — Show weather forecasts for trip dates
18. **Photo Gallery** — Attach photos to completed activities

Idea Prioritization Matrix

Feature	Impact	Effort	Priority
AI Itinerary Generation	High	High	P0 — Critical
User Authentication (Supabase)	High	Medium	P0 — Critical

Feature	Impact	Effort	Priority
Trip CRUD Operations	High	Medium	P0 — Critical
Day-by-Day View	High	Medium	P0 — Critical
Budget Tracking	High	Low	P0 — Critical
Activity Edit/Delete	Medium	Medium	P1 — Important
Day Regeneration	High	Medium	P1 — Important
Drag-and-Drop Reorder	Medium	Medium	P1 — Important
Travel Style Support	Medium	Low	P1 — Important
Interest-Based Personalization	Medium	Low	P1 — Important
Multi-Currency Support	Low	Low	P2 — Nice-to-have
Collaborative Planning (RLS)	Medium	High	P2 — Nice-to-have
Map Integration	Medium	High	P3 — Future
Export to Calendar	Low	Medium	P3 — Future

Selected Approach

After brainstorming and prioritization, the following approach was selected:

Architecture: Full-stack web application with decoupled frontend and backend

- **Frontend**: React 19 + TypeScript + Vite + TailwindCSS v4 + shadcn/ui + React Query
- **Backend**: Python FastAPI with RESTful API design
- **Database**: Supabase (PostgreSQL) with Row-Level Security (RLS)
- **AI Engine**: Google Gemini API (primary) with Groq API (fallback)
- **Deployment**: Vercel (frontend) + Render (backend)

Key Decision: Use Generative AI (LLMs) for itinerary generation rather than rule-based algorithms, enabling truly personalized and context-aware travel plans with real-world places and accurate cost estimates.